# *Chapter 3*
# *Vector Spaces*

## 3.3 Linear Independence

# Linear Dependence

**Definition:**
The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly dependent (线性相关)** if and only if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ ,**not all zero**, such that

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = 0$$

- Let $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n]$, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is linearly dependent if and only if the null space of $A$, $N(A)$, is NOT the zero subspace {**0**}.
- **Theorem**
  Nonzero vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly dependent if and only if at least one vector is a linear combination of the others.

# Linear Independence

**Definition:**

The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly independent (线性无关)** if

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = 0$$

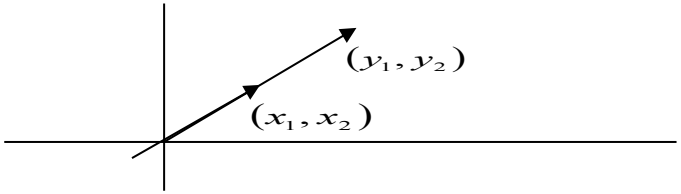
implies that all the scalers $\alpha_1, \alpha_2, \ldots, \alpha_n$ **must equal to** 0.

- Let $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n]$, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is linearly independent if and only if the null space of $A$ IS the zero subspace: $N(A) = \{\mathbf{0}\}$.
- None of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ can be written as a linear combination of the others.

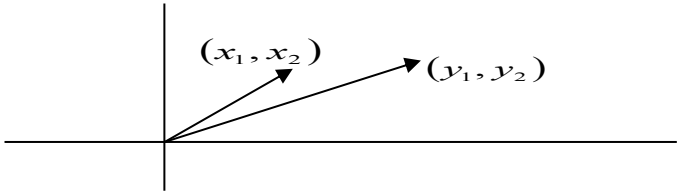
# Quick Example 1

- When is a set of two nonzero vectors linearly dependent?

When one vector can be written as a scaler multiple of the other. Geometrically, if the two vectors are placed at the origin, they will lie along the same line.




**x** and **y** linearly dependent




**x** and **y** linearly independent

- Is a set containing the zero vector linearly dependent or linearly independent?

Linearly dependent. Let the scalers of all other vectors be zero, and the scaler of the zero vector be nonzero.

# Example 2

Are the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}$ in $\mathbb{R}^4$ linearly dependent or independent?

# Example 2

Are the vectors $\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\\2\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}0\\1\\1\\2\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}1\\1\\1\\3\end{bmatrix}$ in $\mathbb{R}^4$ linearly dependent or independent?

**Solution** If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = 0$, solve the homogenous linear system

$$\alpha_1 + \alpha_3 = 0$$
$$\alpha_2 + \alpha_3 = 0$$
$$\alpha_1 + \alpha_2 + \alpha_3 = 0$$
$$2\alpha_1 + 2\alpha_2 + 3\alpha_3 = 0$$

By Gauss-Jordan reduction:

$$\left[\begin{array}{ccc|c}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & 0\end{array}\right]$$

The only solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

Therefore $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ are linearly independent.

# Example 3

Are the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ in $\mathbb{R}^3$ linearly dependent or independent?

# Example 3

Are the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ in $\mathbb{R}^3$ linearly dependent or independent?

**Solution**

If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 + \alpha_4\mathbf{v}_4 = 0$, solve the homogenous linear system

$$\alpha_1 + \alpha_2 - 3\alpha_3 + 2\alpha_4 = 0$$
$$2\alpha_1 - 2\alpha_2 + 2\alpha_3 = 0$$
$$-\alpha_1 + \alpha_2 - \alpha_3 = 0$$

By Gauss-Jordan reduction:

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The solutions of the system are $[\alpha_3 - \alpha_4, 2\alpha_3 - \alpha_4, \alpha_3, \alpha_4]$, so there are nontrivial solutions, therefore the vectors are linearly dependent.

# Example 4

Are the vectors in $P_3$ (polynomial functions)

$$p_1(t) = t^2 + 4t + 14$$

$$p_2(t) = 2t^2 + t$$

linearly dependent or independent?

# Example 4

Are the vectors in $P_3$ (polynomial functions)

$$p_1(t) = t^2 + 4t + 14$$
$$p_2(t) = 2t^2 + t$$

linearly dependent or independent?

**Solution**

To test whether $p_1(t)$ and $p_2(t)$ are linearly independent, set

$$\alpha_1 p_1(t) + \alpha_2 p_2(t) = 0t^2 + 0t + 0$$

We get $(\alpha_1 + 2\alpha_2)t^2 + (4\alpha_1 + \alpha_2)t + 14\alpha_1 = 0$

Equating coefficient lead to the system

$$\alpha_1 + 2\alpha_2 = 0$$
$$4\alpha_1 + \alpha_2 = 0$$
$$14\alpha_1 = 0$$

The only solution is $\alpha_1 = 0, \alpha_2 = 0$.

Therefore $p_1(t)$ and $p_2(t)$ are linearly independent.

# Theorem

## Theorem 3.3.1

Let $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ be n vectors in $\mathbb{R}^n$ and let $X = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$. The vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ will be linearly dependent if and only if $X$ is singular.

**Warning:** This theorem only works with n vectors in $\mathbb{R}^n$ (square matrix case)

*Proof* The equation

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n = \mathbf{0}$$

can be rewritten as a matrix equation

$$X\mathbf{c} = \mathbf{0}$$

This equation will have a nontrivial solution if and only if $X$ is singular. Thus, $\mathbf{x}_1, \ldots, \mathbf{x}_n$ will be linearly dependent if and only if $X$ is singular. ■

We can use Theorem 3.3.1 to test whether $n$ vectors are linearly independent in $\mathbb{R}^n$. Simply form a matrix $X$ whose columns are the vectors being tested. To determine whether $X$ is singular, calculate the value of $\det(X)$. If $\det(X) = 0$, the vectors are linearly dependent. If $\det(X) \neq 0$, the vectors are linearly independent.

**Theorem 3.3.2**

Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be vectors in a vector space $V$. A vector $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ can be written *uniquely* as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$ if and only if $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent.